

3 Chemicals are used to regulate responses in both plants and animals.

(a) In mammals antidiuretic hormone (ADH) is involved in regulating plasma concentration and blood volume.

(i) Which of the following produces ADH?

(1)

- ☐ **A** adrenal gland
- ☐ **B** hypothalamus
- ☐ **C** medulla oblongata
- ☐ **D** pituitary gland

(ii) Which structures does ADH act on?

(1)

- ☐ **A** capillaries
- ☐ **B** collecting ducts
- ☐ **C** proximal tubules
- ☐ **D** ureters

(iii) The control of plasma concentration and blood volume in the body is an example of which type of process?

(1)

- ☐ **A** countercurrent multiplier
- ☐ **B** habituation
- ☐ **C** negative feedback
- ☐ **D** positive feedback

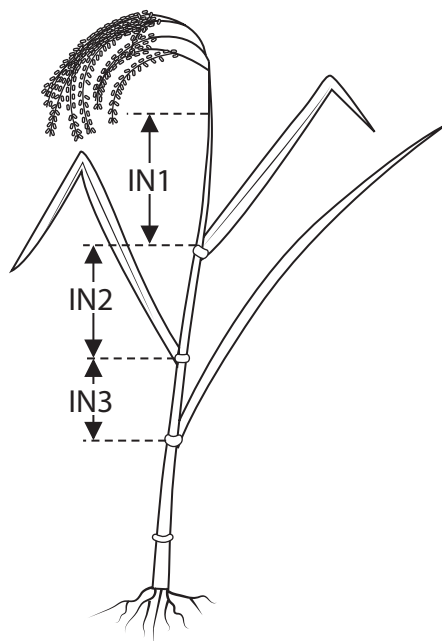
(iv) Name the type of receptor in the body that detects changes in the plasma concentration.

(1)

(b) The effect of a gibberellin on the growth of cereal plants was investigated.

Different concentrations of the gibberellin were sprayed on cereal plants.

The lengths of the first, second and third internodes (IN1, IN2, IN3) were measured, as shown in the diagram.



The table shows the effect of the gibberellin on the mean lengths of the first, second and third internodes.

Concentration of the gibberellin / ppm	Mean length / cm		
	first internode	second internode	third internode
0	36	19	14
10	36	19	14
100	41	21	14
1000	52	22	15
2500	56	22	15

(i) Comment on the effect of the gibberellin on the growth of cereal plants.

(2)

(ii) Suggest how the gibberellin causes this effect.

(3)

(Total for Question 3 = 9 marks)